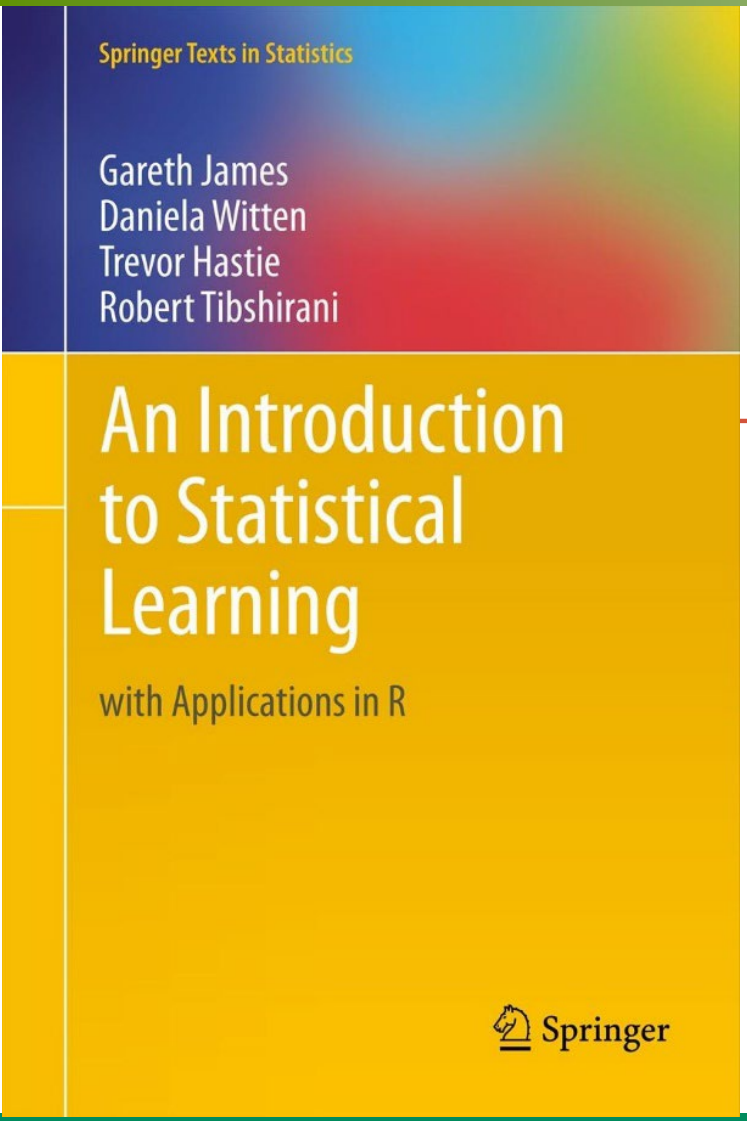


ANALYTICS

Week 5

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Examples Inspired By CAS, SOA, Coaching Actuaries ©

Key Formulas This Week

- $se(b_j) = s\sqrt{(j+1)st \text{ diagonal of } (\boldsymbol{X'X})^{-1}}$
- $Cov(b_i, b_j) = s^2[(i+1, j+1)st \text{ element of } (\boldsymbol{X'X})^{-1}]$
- $t(b_j) = \frac{b_j - H_0(b_j)}{se(b_j)}$ $w/df = n - (p+1)$
- $r_{partial} = \frac{t_i}{\sqrt{t_i^2 + n - (p+1)}}$
- CI/PI: Estimate $\pm$ t-Critical Value * Standard Error
- $F = \frac{\frac{SSReg}{p}}{\frac{SSE}{n-(p+1)}} = \frac{\frac{R^2}{p}}{\frac{1-R^2}{n-(p+1)}}$ or $F = \frac{\frac{(SSE_0 - SSE_{full})}{q}}{\frac{SSE_{full}}{n-(p+1)}} = \frac{\frac{R^2_{full} - R^2_{reduced}}{q}}{\frac{1-R^2_{full}}{n-(p+1)}}$

Coefficient Interpretation

```
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.3863     0.1685  -8.225  < 2e-16 ***
ALow          0.7133     0.2980   2.394   0.0167 *
BLow         -1.3940     0.2115  -6.590 4.41e-11 ***
ALow:BLow     1.4142     0.3467   4.079 4.53e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

- ANormal, BNormal are the base categories.

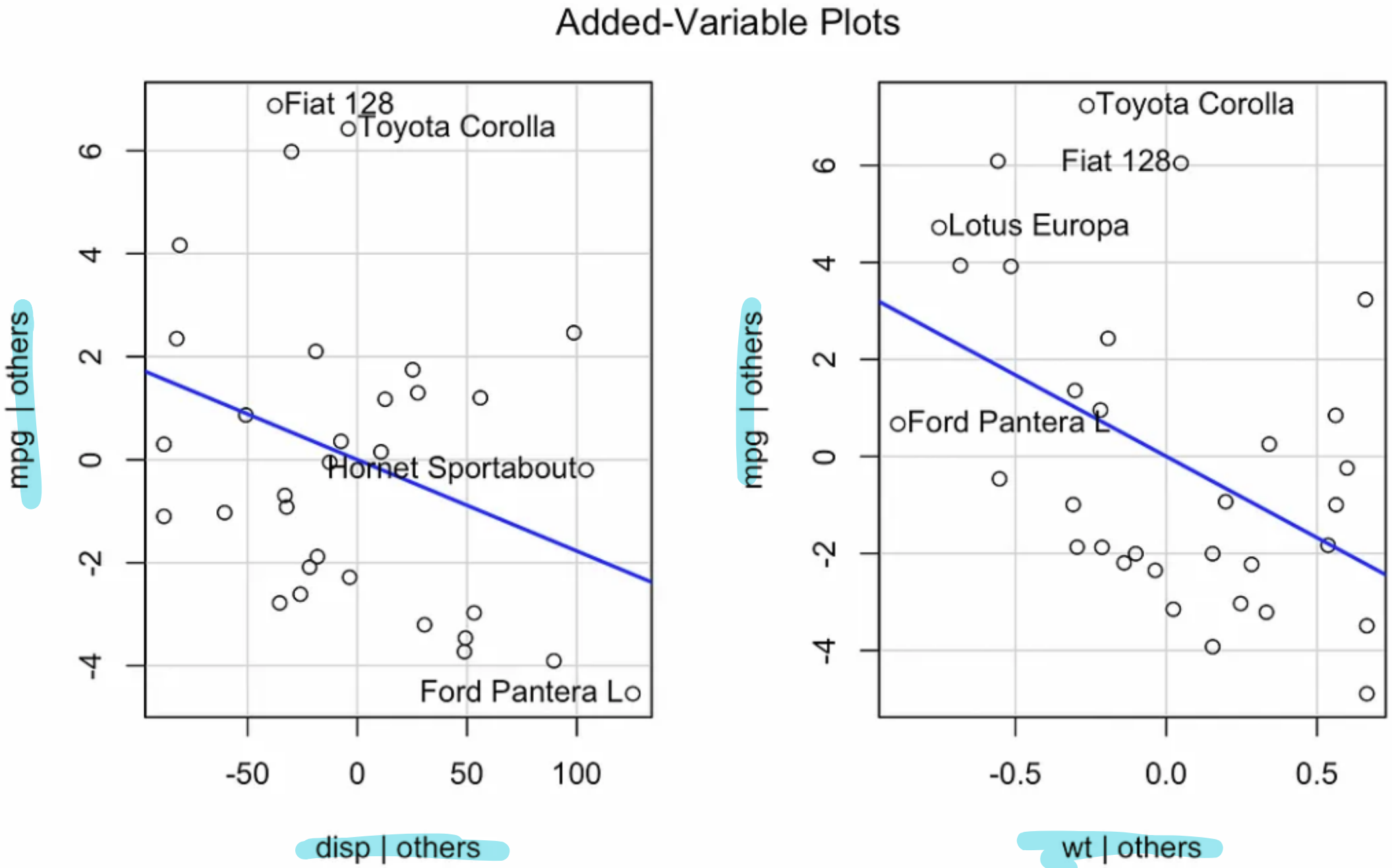
Parameter SEs

- Standard Error = $\sqrt{\text{Variance}}$
- $se(b_j) = s\sqrt{(j+1)st \text{ diagonal of } (\boldsymbol{X'X})^{-1}}$
- $Cov(b_i, b_j) = s^2[(i+1, j+1)st \text{ element of } (\boldsymbol{X'X})^{-1}]$

Added Variable Plots

- **Added variable plots** (partial regression) try to isolate the effect of adding a predictor x_{p+1} to a model.
- 1. Run a regression $y \sim x_1 + x_2 + \cdots x_p$ and store the residuals e_y.
- 2. Run a regression $x_{p+1} \sim x_1 + x_2 + \cdots x_p$ and store the residuals e_x.
- 3. Plot e_y against e_x.

Added Variable Plot



Partial Correlation

- The **partial correlation coefficient** is the correlation between e_x & e_y in a partial variable plot
- $r_{partial} = \dfrac{t_i}{\sqrt{t_i^2 + n - (p+1)}}$
  - p: Total predictors including the ith
- $t_i = \dfrac{b_i}{se(b_i)}$

Predictors Needed *

- The # of explanatory predictors needed quantifies model flexibility.
- Each continuous explanatory variable adds 1 predictor.
- Each categorical explanatory variable adds $m - 1$ predictors.
- Continuous w/continuous interaction term adds 1 predictor.
- Continuous w/categorical interaction term adds $m - 1$ predictors.
- n-level categorical w/m-level categorical interaction term adds $(n - 1)(m - 1)$ predictors

1. # Predictors Illustration

- You are given a multiple linear regression model of claim size with the following potential explanatory variables only: +1 intercept
- House value, a continuous variable modeled with a 4th degree polynomial (4)
- Owner age, a continuous variable modeled linearly (1)
- Owner Gender, a categorical variable with 2 levels (2-1) (1)
- # of occupants, a categorical variable with 5 levels (5-1) (4)
- One interaction term, between gender and # of occupants. (2-1)(5-1) (4)
- Determine the max # of parameters in this model. 15

t-Test

- Statistical hypothesis tests help determine whether explanatory variables x_j are important by testing $H_0: \beta_j = 0$ vs. $H_a: \beta_j \neq 0$.
- Establish a significance level α.
- $t(b_j) = \frac{b_j - H_0(b_j)}{se(b_j)}, df = n - (p+1)$ if assumptions 1-5 hold
- Reject H_0 if $|t(b_j)| > t_{n-(p+1),\frac{\alpha}{2}}$ for 2-sided
- Or, just use the p-value output directly.

t-Test Summary *

- Small $p \rightarrow$ Higher magnitude t-statistic, more evidence to reject $H_0 \rightarrow$ Keep parameter in model (it's significant)
- Large $p \rightarrow$ Smaller magnitude t-statistic, fail to reject $H_0 \rightarrow$ Drop parameter from model (confidence interval overlaps 0)
- The pre-set standard for rejecting H_0 is α.

Example 1

- You wish to explain y (Math score) using a MLR model w/16 observations:
- $\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + b_4 x_4$
- Sleep has 3 levels: Low, Medium (Base), High
- For each explanatory variable, you decide to test the null hypothesis $H_0: b_i = 0$ at $\alpha = 10\%$ significance.
- A. Which variable can be removed without data adjustment?
- B. What is the partial correlation coefficient between Math score and Hours?

	Coefficient	Standard Error	t
$\beta_1 = SleepLow$	-3	1	-3
$\beta_2 = SleepHigh$	5	5	1
$\beta_3 = Hours$	8	3.5	2.29
$\beta_4 = \#Coffees$	9	6	1.5

$df = 16 - (4+1) = 11$ Crit $t_{11,10\%} = 1.796$

drop coffees

Example 1 Sol

- $t(b_j) = \frac{b_j}{se(b_j)}, df = n - (p+1)$
- $r_{partial} = \frac{t_i}{\sqrt{t_i^2 + n - (p+1)}} = \frac{2.29}{\sqrt{2.29^2 + 16 - (4+1)}}$

$= .58$ or $.57$ w/ more decimals)

Confidence Intervals

- Confidence intervals provide a range of reliability for each parameter estimate, given a chosen significance level α.
- CI: $b_j \pm t_{critical} * se(b_j)$
- Estimate $\pm$ (Crit) · (Std. error)

Prediction Intervals

- Prediction intervals provide a range of reliability for each $\hat{y}_{pred}$ from a model, and are wider due to model uncertainty + irreducible error

- FYI PI: $\hat{y}_{pred} \pm t_{critical}\sqrt{s^2(1 + \boldsymbol{x}_*^T(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{x}_*^T)}$
  - $\boldsymbol{x}_*^T$: Vector of new inputs for prediction

Prediction Intervals 2

- Prediction accuracy is limited by:
- Reducible error (inaccuracy in b coefficient estimates)
- Model bias (linear model ≠ reality)
- Irreducible error (random error)
- Use prediction intervals for individual predictions, confidence intervals for means, parameters estimates.

2. CI Illustration

- Consider the following dataset. You fit a regression model to y using two explanatory variables, x_1 and x_2.
- From the fitted model, root mean squared error is 1.373.
- $b = \begin{pmatrix} 0.15 \\ 0.69 \\ 2.88 \end{pmatrix}, (\boldsymbol{X'X})^{-1} = \begin{pmatrix} 0.54 & -0.077 & -0.15 \\ -0.077 & 0.154 & -0.69 \\ -0.15 & -0.69 & 4.12 \end{pmatrix}$ s
- Determine the 95% confidence interval for b_2.

i	1	2	3	4
$x_{i,1}$	-1	2	4	6
$x_{i,2}$	0	0	1	1
y_i	0	1	5	8

Var-Cov Mat $= s^2 (X'X)^{-1}$

$se(b_2) = 1.373 \sqrt{4.12} = 2.787$

95% CI: $2.88 \pm 12.7062 \cdot 2.787$

$= (-32.53, 38.29)$

t_{crit}, 5% 2-sided

$df = 4 - (2+1) = 1$

2. CI Illustration 2

- CI: $b_j \pm t_{critical} * se(b_j)$
- $se(b_j) = s\sqrt{(j+1)st \text{ diagonal of } (\boldsymbol{X'X})^{-1}}$

Example 2

- A multiple linear regression model built on 3 explanatory variables and 18 observations has a 95% confidence interval for b_2 of $(-3.6, -1)$.
- Find the t-statistic for testing $H_0: b_2 = 0$.
- Is the explanatory variable associated with b_2 significant? Should it be kept or removed from the model?

$b_2 = \frac{-3.6 - 1}{2} = -2.3$

$t = \frac{b_2}{se(b_2)} = \frac{-2.3}{.6061} = -3.795$

$-1 = -2.3 + t \cdot se(b_2)$

$t \sim 2.1448$

crit $df = 18 - (3+1) = 14$

$se(b_2) = .6061$

Example 2 Sol

- CI: $b_j \pm t_{critical} * se(b_j)$
- $t(b_j) = \frac{b_j}{se(b_j)}, df = n - (p + 1)$

ANOVA

- The **ANOVA** (analysis of variance) table summarizes the sources of variability.
- Mean square is the sum of square values divided by its respective degrees of freedom.

Source	Sum of Squares	df	Mean Square
Regression	$SSReg$	p	$MSR = \frac{SSReg}{p}$
Error	SSE	$n-(p+1)$	$MSE = \frac{SSE}{n-(p+1)}$
Total	SST	$n-1$	$= s_y^2$

2 F-test Versions

- 1. Determine significance of an entire model or
- 2. Test whether a more complex model is better than a simpler nested version
- A predictor can be significant by a t-test while a whole model F-test is not significant because they test different hypotheses.

Whole F-test

- To test an entire model, H_0: $\beta_1 = \beta_2 = \cdots \beta_p = 0$
- H_a: At least one β_j is non-zero
- $F_{p,n-(p+1)} = \frac{MSReg}{MSE} = \frac{\frac{SSReg}{p}}{\frac{SSE}{n-(p+1)}} = \frac{R^2}{1-R^2} \cdot \frac{n-(p+1)}{p}$
  - p: # of explanatory variables (excludes fitted intercept)
  - R^2: of full model

SLR F-test

- For basic linear regression, the F-statistic is directly related to the t-statistic:

- $t = \sqrt{F}$

- Higher $MSReg$ relative to MSE is more evidence to reject H_0.

Nested F-test

- To compare whether a subset model is better,
- H_0 (reduced): A subset q of the β_i coefficients are 0
- H_a (full): At least one of the β_i in the subset is non-zero
- $F = \dfrac{\frac{(SSE_0 - SSE_{alt})}{q}}{\frac{SSE_{alt}}{n-(p+1)}} = \dfrac{\frac{(R^2_{alt} - R^2_0)}{q}}{\frac{1-R^2_{alt}}{n-(p+1)}}$
- $t = \sqrt{F}$ if 1 predictor is omitted for H_0 vs. H_a.

3. F-test Illustration (SOA)

- A. For a simple linear regression model with 8 observations, if $\sum_{i=1}^{8}(y_i - \hat{y}_i)^2 = 200$ and the F statistic is 3, calculate the coefficient of determination.
- B. For the multiple regression model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \epsilon$ to estimate house size, you are given:
- The # of observations is 3,120.
- The total sum of squares is 15,000.
- $H_0: \beta_3 = \beta_4 = \beta_5 = 0$
- The coefficient of determination for the full model is 40%.
- The regression sum of squares for the reduced model is 5,500.
- Determine the value of the F-statistic for testing H_0.

(A) $3 = \dfrac{R^2}{1-R^2} \cdot (8-2) \rightarrow R^2 = \dfrac{1}{3}$

$SSReg = 100$

$3 = \dfrac{MSReg}{MSE} = \dfrac{SSReg}{\frac{SSE}{n-2}}$ ($SSE \rightarrow 200$, $8 \leftarrow n-2$)

$R_A^2 = \dfrac{100 \rightarrow SSReg}{\underbrace{100+200}_{SST}} = \dfrac{1}{3}$

(B) $F = \dfrac{\frac{R_a^2 - R_0^2}{q \rightarrow 3} \rightarrow .3\bar{6}}{\frac{1-R_a^2}{n-(p+1)}} = 57.\bar{6}$

$R_0^2 = \dfrac{SSReg_0}{SST} = \dfrac{5500}{15{,}000} = .3\bar{6}$

$F = \dfrac{\frac{SSE_0 - SSE_a \rightarrow 9{,}000}{q \;\; 33}}{\frac{SSE_a \rightarrow 9{,}000}{n-(p+1) \;\; 3120-(5+1)}} = 57.\bar{6}$

$9{,}500 \leftarrow SSE_0 - SSE_a$

$SSE_0 = SST - SSReg_0 = 15{,}000 - 5{,}500 = 9{,}500$

$3{,}114$

$$.4 = \frac{SST - SSE}{SST} = \frac{15{,}000 - SSE_a}{15{,}000}$$

$$SSE_a = 9{,}000$$

3. F-test Illustration

- SLR: $F = \frac{MSReg}{MSE} = (n-2)\frac{R^2}{1-R^2}$
- $R^2 = \frac{SSR}{SST} = \frac{SST-SSE}{SST}$
- $F = \frac{\frac{(SSE_0 - SSE_{alt})}{q}}{\frac{SSE_{alt}}{n-(p+1)}} = \frac{\frac{R^2_{alt} - R^2_0}{q}}{\frac{1-R^2_{alt}}{n-(p+1)}}$

Example 3 (SOA)

- Twenty pairs of observations (x_i, y_i) are fitted to a linear regression model A: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$.
- $\sum x_i = 100$
- $\sum y_i = 200$
- $\sum x_i^2 = 2{,}100$
- $\sum y_i^2 = 8{,}000$
- $b_1 = 1.1$
- The F-ratio for testing $H_0: \beta_1 = 0$ in model A equals the F-ratio for testing $H_0: \beta_1 = \beta_2 = 0$ in model B on the same data with an extra predictor x_{2i}: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$.
- What must be the regression sum of squares for Model B?

$F_A = \dfrac{SSReg_A}{SSE_A / (n-2)} \rightarrow 1{,}936$

$SSReg_A = 1.1^2\left(2100 - \dfrac{100^2}{20}\right) = 1{,}936$

$SSE_A = 6000 - 1936$

$20 \leftarrow n-2$

$SST = 8000 - \dfrac{200^2}{20} = 6{,}000$

$F_A = 8.5748 =$ $\frac{R^2}{1-R^2} \cdot \frac{20-(2+1)}{2} = F_B$

$R_B^2 =$ $= \frac{SSReg_B}{6,000} = .502$

$SSReg_B = 3,013$

$\frac{R^2}{1-R^2} = 1.0088$

$2.00880^2 = 1.0088$ $\quad R^2 = .502 =$

or $8.5748 = \frac{\frac{SSReg_B}{2}}{\frac{6000 - SSReg_B}{20-(2+1)}}$

$SSReg_B = 3,013$

Example 3 Sol

- $F_{p,n-(p+1)} = \frac{MSReg}{MSE} = \frac{\frac{SSReg}{p}}{\frac{SSE}{n-(p+1)}} = \frac{R^2}{1-R^2} * \frac{n-(p+1)}{p}$
- $SST = \sum(y_i - \bar{y})^2 = \sum y_i^2 - \frac{(\sum y_i)^2}{n}$
- $SSReg = \sum(\hat{y}_i - \bar{y})^2 = b_1^2\left(\sum x_i^2 - \frac{(\sum x_i)^2}{n}\right)$ in SLR

Summary Check

- *Can you describe in your own words:*
- How an added variable plot works and the partial correlation formula
- How to count number of predictors
- How to conduct a t-test
- How a full-model F-test vs. partial F-test work and their formulas
- How to find confidence and prediction intervals